Ernest Mlati Mwang'ombe

Backend AI Engineer | IT Systems Administrator | AI Quality Analyst

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Professional Summary

A Backend AI Engineer and IT Systems Administrator with over 10 years of experience spanning cloud infrastructure, identity systems, and AI backend engineering. Documented history of architecting Node.js/Express REST microservices, parameterised relational database schemas, and multi-model LLM evaluation frameworks. Combines hands-on experience in enterprise systems security (Microsoft Entra ID, Active Directory, M365) with technical expertise in API design, type-safe data pipelines, and AI model alignment.

Core Competencies

- **Backend AI & API Engineering:** Node.js, Express.js, REST microservices, OpenAPI 3.0 (Swagger), Zod runtime validation, automated web scraping (Cheerio).
- **Database & Security Pipelines:** Relational schema design (SQLite), parameterised SQL queries, Supabase Auth, JWT verification middleware, API security.
- **AI Evaluation & Prompt Engineering:** LLM grounding, hallucination auditing, multi-model benchmarking, 5-stage prompt ladders, SxS rating rationales.
- **Identity & Access Management:** Microsoft Entra ID (Azure AD), Active Directory, Identity Lifecycle Management, RBAC, least privilege, MFA, SSO, PIM.
- **Systems & Cloud Administration:** Microsoft 365, Google Workspace, Windows Server, Exchange Online, patch management, disaster recovery, Citrix VDI, PowerShell scripting.
- **Network Infrastructure & Cybersecurity:** Router/switch configuration (Cisco, Netgear, Ubiquiti), TCP/IP, DNS, DHCP, LAN/WAN, endpoint protection, GDPR compliance.

Professional Experience

Backend AI Engineer Jun 2026 – Present
FlyRank (Remote)

- Architected RESTful microservices using Node.js and Express to handle core API workflows, incorporating input validation middleware and OpenAPI 3.0 documentation.
- Engineered disk-backed relational database schemas in SQLite using parameterised SQL queries to eliminate injection vulnerabilities and maintain data persistence.
- Built authentication and authorisation pipelines leveraging Supabase Auth and custom JWT verification middleware to safeguard protected backend endpoints.
- Developed automated web scraping pipelines with Cheerio and Zod runtime schema validation to convert unstructured web data into type-safe JSON formats.
- Designed multi-model evaluation frameworks and prompt ladders to benchmark response quality, token usage, and logical adherence across LLM architectures.

IT Consultant May 2018 – Present
Elike Trade Mart (Kenya)

- Lead technical support and systems engineering for 500+ employees, managing Google Workspace and Microsoft Entra ID.
- Achieved a 20% reduction in average ticket response and resolution times by introducing diagnostic frameworks for hardware, software, and network failures.
- Designed and maintained an internal Knowledge Base, reducing routine support tickets by approximately 15% and streamlining user onboarding.
- Spearheaded training initiatives for software adoptions, boosting system utilisation rates by 30% across more than 2,500 interactions.

IT Network Technician Aug 2020 – Nov 2023
Darnov Technology (Kenya)

- Maintained Windows Server environments, including Active Directory, DNS, DHCP, and secure file hosting, ensuring 99.9% uptime for business applications.
- Reduced network incidents by 25% by deploying network monitoring systems (Ubiquiti, Netgear, Cisco) and conducting scheduled firmware and security patching.
- Developed PowerShell automation scripts to handle routine administrative tasks, reducing manual system overhead.

AI Data Quality Analyst & LLM Annotator Oct 2021 – Mar 2024
Remotask (Remote)

- Executed evaluation, labelling, and validation processes for AI initiatives, specialising in multi-turn LLM safety testing and grounding.
- Conducted audits of LLM outputs to identify, catalogue, and resolve hallucinations, ensuring responses adhered to instructional logic and safety guidelines.
- Developed multi-turn prompt scenarios to test boundary limits and evaluate AI performance under edge-case parameters.
- Authored structured rationales detailing reasoning behind SxS evaluation ratings to support model alignment loops.

IT Administrator (Contractor) Jan 2017 – Dec 2017
Hess Corporation / Collabera (Houston, TX / Hybrid)

- Supported an enterprise environment of over 15,000 users operating on Windows platforms, maintaining strict SLA metrics.
- Led an Active Directory clean-up project, auditing and removing over 300,000 obsolete accounts and resources to achieve a documented 15% reduction in licensing costs while improving directory security.
- Configured and deployed Group Policy Objects (GPOs) to enforce security policies and standardise user profiles across divisions.
- Managed Exchange Online user resources, distribution listings, and migrations from on-premises Exchange architectures to Microsoft 365.
- Evaluated third-party software solutions to ensure alignment with internal security policies before deployment.

Customer Engineer (Contractor) Jul 2016 – Jan 2017
NCR Corporation / Shoening Technology (Houston, TX)

- Maintained 98% uptime across regional banking ATM hardware and software clusters by delivering calibration and troubleshooting.
- Managed an average of 15 on-site service calls per week, exceeding target KPIs with a 95% first-visit resolution rate.
- Executed ATM operating system upgrades from Windows XP to Windows 7, verifying hardware compatibility, software integrity, and transactional security.
- Upgraded legacy magnetic stripe card readers with EMV chip card readers and software systems to satisfy regulatory compliance standards.

IT Help Desk Administrator (Contractor) Feb 2013 – Jun 2016
J.P. Morgan / Collabera (Houston, TX / Hybrid)

- Contributed to an improvement in Tier-2 ticket resolution rates, elevating average SLA compliance from 70% to 98% through targeted troubleshooting and workflow optimisations.
- Assisted over 15,000 users across desktops and virtualised environments (Citrix VDI), managing and debugging remote sessions.
- Configured User Access Control, RBAC permissions, and directory structures to ensure compliance with financial and data privacy standards (including PCI DSS).
- Administered multi-factor authentication (MFA) tokens using the RSA Administration Tool, supporting secure networks.
- Managed mainframe access frameworks (including Rumba, AS400, TSO, ACF) and configured Virtual Telecommunications Access Method (VTAM) profiles.
- Investigated security escalations, documented exposure vectors, and coordinated directly with threat response units.

Education

Associate of Science in Computer Systems, Networking & Telecommunications
Lone Star College, Houston, Texas, USA (2010 – 2012)
Credential Records: Europass E-Portfolio Verified Record | Digital Test Results

Professional Certifications

1. AI Fluency: Framework & Foundations (Anthropic Education) | Verification Link
2. Introduction to Modern AI & Chatbots (Credly) | Credly Badge
3. Getting Started with Google Gemini (LinkedIn Learning) | LinkedIn Certificate
4. Web Design with AI (Mayerfeld Consulting) | Verification Link
5. Cisco Cybersecurity Pathway | Credly Badge
6. Microsoft Entra ID (Azure AD) | LinkedIn Certificate
7. Zendesk Administration & Support | LinkedIn Certificate
8. Customer Service: Problem-Solving & Troubleshooting | LinkedIn Certificate
9. EF SET English Certificate (C2 Proficient) | EF SET Verification

Languages

- **English:** Proficient (C2 CEFR Certification) — Primary Business & Technical Language
- **Swahili:** Native Speaker — Fluent Verbal & Written